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“Enabling the Internet of Things in an Automobile”

We are nearing the era where your car becomes smart enough to cool the interiors before you arrive, alert you if you have forgotten something, and respond to your instructions from across the world. This interview talks about the challenges in implementing such next-gen features in an electric car.

Chetan Maini, founder, chief of technology and strategy, Mahindra Reva, speaks to Dilin Anand and Ashwin Gopinath of EFY about Internet of Things (IoT)-enabled cars like the new *Mahindra-Reva e2o*, and the challenges in implementing the multitude of technologies that have gone into it

The new e2o seems like a true IoT car. Could you talk about how you went about taking connectivity to the next level in it?

The e2o was the result of answering a lot of questions like, ‘Why does my car need fossil fuels, and why can’t it use the sun?’, as well as ‘Why can’t my car talk to me and let me know when something is wrong?’ I can easily say that looking at some of these questions from a very fundamental angle gave us most of our solutions, i.e., the functionalities offered in the new car. Our engineers did the work, and from their feedback came the new management system, the new battery system, new ways to put the car together, and the usage of light-weight materials and plastics. Now, with over ten on-board computers that control the entire vehicle, we have come a long way to developing a proper and affordable electric vehicle (EV) that will make a meaningful difference to people’s lives.

The e2o is online at all times; consumers have applications on their iPhone, Android and BlackBerry devices, through which they can control the air-conditioning of the car, check the status of the car’s vitals, and map their routes according to live traffic updates to save time. Also, the level of connectivity is so high that, soon, we will be able to perform diagnostics on the car, remotely enabling the car’s servicing to be done online.

What kind of electronics hardware has gone into the e2o to enable this shift to a connected car?

The e2o has ten ECUs, all of which are connected on a CAN bus system and talk to each other. The main system is the energy management system, which has approximately 120 I/Os—it manages every battery cell, the temperature, voltage and current along with managing the entire battery ecosystem. It also talks to all the other computers on board, thus exchanging information with the cluster, the infotainment section, the telematics device, the HVAC (heating, ventilation, and air-cooling) section of the car, the main motor control, and so on.

The other ECUs include the telematics module, which is similar to your phone with a GSM chipset in it. You also have a smart cluster, which takes in all the information from outside and displays it effectively for the driver’s convenience. We have a very smart car infotainment system, running on the Windows CE platform, which takes care of all the usual entertainment needs along with connecting with the CAN bus system to tell you how far you have driven, how much further you need to drive, how you have driven over the last few days, how much petrol you have saved, and so on.

Then you have the regular ECUs like the body control module. We have a keyless entry feature in the car, so there